

FLIGHT RECORDER

The Fossil Record of Artificial Cognition

Purpose

The **Flight Recorder** captures every evolutionary event in WAFT. Like a black box on an aircraft, it provides complete telemetry for understanding what happened and why.

Why It Matters

WAFT produces scientific data. The Flight Recorder ensures that data is complete, traceable, and reproducible. Future researchers can study the phylogenetic history of any agent lineage.

Event Types

Event	Description
SPAWN	New agent created (generation 0)
MUTATE	Agent genome modified
GYM_EVAL	Agent evaluated in Scint Gym
BREED	Two agents combined to create offspring
DEATH	Agent removed from population (fitness < threshold)
CHECKPOINT	Periodic state snapshot
EMERGENCE	Potential God-Head behavior detected

Event Schema

Every event follows a consistent structure:

```
1  {
2    "event_id": "evt_abc123...",
3    "timestamp": "2026-01-20T00:15:30Z",
4    "event_type": "GYM_EVAL",
5    "genome_id": "sha256:def456...",
6    "parent_id": "sha256:ghi789...",
7    "generation": 5,
8    "fitness": 0.73,
```

JSON

```
9   "metadata": {
10     "scints_encountered": 10,
11     "scints_stabilized": 7,
12     "tokens_used": 4521,
13     "duration_ms": 12340
14   }
15 }
```

Querying the Flight Recorder

List Recent Events

```
1 waft flight list --limit 20
```

[Shell](#)

Filter by Event Type

```
1 waft flight query --type GYM_EVAL --since "1 day ago"
```

[Shell](#)

Filter by Agent

```
1 waft flight query --genome sha256:abc123...
```

[Shell](#)

Show Full Event Details

```
1 waft flight show evt_abc123...
```

[Shell](#)

Lineage Tracing

View Agent Ancestry

```
1 waft flight lineage sha256:abc123... --depth 10
```

[Shell](#)

Output:

```
1 Generation 0: sha256:root... (SPAWN)
2   └─ Generation 1: sha256:mut1... (MUTATE, fitness: 0.52)
3     └─ Generation 2: sha256:mut2... (MUTATE, fitness: 0.61)
4       └─ Generation 3: sha256:bred... (BREED, fitness: 0.68)
5         └─ Generation 4: sha256:mut3... (MUTATE, fitness: 0.71)
6           └─ Generation 5: sha256:abc123... (current, fitness: 0.73)
```

Compare Lineages

```
1 waft flight compare sha256:abc... sha256:def...
```

[Shell](#)

Data Export

JSON Export

```
1 waft flight export --format json --output events.json
```

[Shell](#)

CSV Export

```
1 waft flight export --format csv --output events.csv
```

[Shell](#)

Phylogenetic Tree

```
1 waft flight tree --format newick --output tree.nwk
```

[Shell](#)

Analysis Tools

Fitness Over Time

```
1 from waft.flight import FlightRecorder
```

[Python](#)

```
2
3 recorder = FlightRecorder()
4 events = recorder.query(
5     event_type="GYM_EVAL",
6     since="7 days ago",
7 )
8
9 # Plot fitness trajectory
10 import matplotlib.pyplot as plt
11 generations = [e.generation for e in events]
12 fitness = [e.fitness for e in events]
13 plt.scatter(generations, fitness)
14 plt.xlabel("Generation")
15 plt.ylabel("Fitness")
16 plt.show()
```

Mutation Impact Analysis

```
1 # Find which mutations improved fitness
2 improvements = recorder.find_improvements(
3     min_delta=0.05, # At least 5% improvement
4 )
5
6 for imp in improvements:
7     print(f"Mutation at gen {imp.generation}:")
8     print(f"  Before: {imp.parent_fitness:.2f}")
9     print(f"  After: {imp.child_fitness:.2f}")
10    print(f"  Delta: +{imp.delta:.2f}")
```

 Python

FLIGHT RECORDER | Every Evolution, Recorded